Project Status & Next Steps

Snapshot of the working tree and what to pick up next

Snapshot

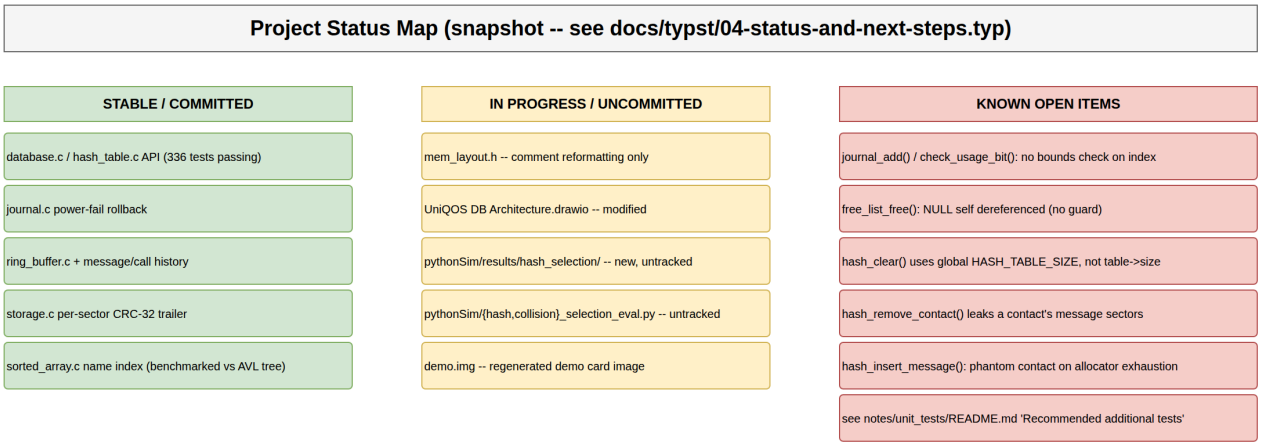


Figure 1: What's stable, what's mid-flight in the working tree, and the known open items – condensed from the sections below.

Working tree, as of this writing

Status	Path	What changed
in progress	`source/app/include/mem_layout.h`	Comment style only – replaced a boxed banner comment with a single-line one. No functional change in the diff seen at generation time.
in progress	`diagrams/UniQOS DB Architecture.drawio`	Modified, uncommitted. 25 pages. Not redrawn by this handoff set; its content wasn't diffed here – check `git diff` before assuming what changed.
in progress	`demo.img`	Regenerated demo SD-card image (binary, used by `source/demo/db_cli`).

open	`docs/`	This handoff set – new, untracked at the time of writing.
open	`pythonSim/hash_selection_eval.py`, `collision_selection_eval.py`	New design-evidence scripts, untracked – see **03 – Benchmarking & Design Evidence**.
open	`pythonSim/results/hash_selection/`	Output of the two scripts above: 25 CSV/PNG/SVG files, untracked.

> None of the untracked `pythonSim` work or this `docs/` set has been committed. If you pick this project up, deciding whether/how to commit them is one of the first calls to make – they're real, finished design evidence, not scratch output.

Known unresolved bug: code 7 (`T_MESSAGE_FIND`)

`notes/code7_message_reconstruction_bug.md` documents a GDB session against real hardware investigating a persistent code-7 failure in `db_run_write()`. Status as recorded there:

- A `message.c` bitmap offset bug (§1) – **fixed, confirmed working**.
- A `hash_reconstruct_message()` scan-bounds bug (§2) – **real bug, a partial fix was attempted and reverted**. Even with that fix applied, code 7 persisted and `insert_message_from_sector` was never called, so **there is at least one more bug in this path not yet found**.

This is the single most concrete lead for whoever picks this up next – read that note before touching message reconstruction.

Stable / committed

The core API (`database.c`, `hash_table.c`) passes all 336 host tests. `journal.c` rollback, `ring_buffer.c` (message/call history), `storage.c`'s per-sector CRC-32 trailer, and `sorted_array.c` (chosen over an AVL tree with benchmark evidence) are all exercised by the test suite and considered settled.

Open items from the test-suite review

`notes/unit_tests/README.md` keeps a prioritised "Recommended additional tests" list; condensed here, see that file for full detail and code references:

1. **No bounds check** on `journal_add()` / `check_usage_bit()` for an out-of-range sector index – `journal_add(&journal, JRNL_CONTACT, UINT16_MAX, content)` segfaults today.
2. `free_list_free()` dereferences a `NULL self` with no guard (unlike `free_list_allocate()`, which checks).

3. `hash_clear()` uses the global `HASH_TABLE_SIZE` instead of `table->size` – a table initialised smaller than the default with a minimally-sized backing array would overrun on clear.
4. Two characterised-but-unresolved design gaps, pinned down by tests but not yet decided on:
   - `hash_remove_contact()` leaks a contact's message sectors if it has any.
   - `hash_insert_message()` leaves a phantom empty contact behind when message-sector allocation fails right after a brand-new contact was committed.
5. Ring buffer iterator off-by-one decisions (upper limit `occupancy` vs. `occupancy - 1`; whether `move_prev` wraps like `move_next`).
6. No death test / known-vector coverage for `crc.c` against published CRC-32 test vectors.
7. Four `.cpp` files under `tests/` aren't wired into `tests/CMakeLists.txt` and don't run: `test_hash_table_messages.cpp`, `test_message_extent.cpp`, `test_my_logic.cpp`, `test_database.cpp` (empty) – worth deciding whether to revive, rewrite, or delete.

Suggested order of attack

1. Read `notes/code7_message_reconstruction_bug.md` and resolve the remaining message-reconstruction bug – it's the only item here blocking correctness rather than coverage.
2. Decide what to do with the uncommitted `pythonSim` design-evidence scripts and this `docs/` set (commit, or fold into the thesis write-up directly).
3. Work through the bounds-check / NULL-guard items (1–3 above) – small, well-isolated fixes with an existing characterisation test pattern to follow.
4. Decide the two characterised design gaps (contact message-sector leak, phantom contact on allocator exhaustion) – these need a design call, not just a test.
5. Revisit the ring buffer off-by-one questions and the four orphaned test files.